Simulation of a Cold-Chain Logistics Monitoring System Using OMNeT++

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I. Introduction and Project Description

Cold-chain logistics play a critical role in transporting temperature-sensitive goods such as medical supplies or perishable food. Any deviation from the needed temperature ranges may result in product expiration or product becomes unsuitable for usage.

This project proposes a simulation-based Cold-Chain Logistics Monitoring System using IoT-enabled sensors embedded in refrigerated containers. These sensors periodically monitor temperature, humidity and mobility, and transmit data to a central backend system using wireless communication.

The primary motivation is to ensure the integrity of sensitive goods by generating urgent alerts when environmental conditions deviate from safe thresholds. The simulation models mobility, intermittent reporting, stochastic behaviour, and communication delays. The goal is to evaluate system performance in terms of reliability and responsiveness.

II. System Architecture and Technical Illustration

Fig. 1 describes the Cold-Chain Logistics Monitoring System. The system consists of sensor nodes which are IoT devices that gather temperature, humidity and mobility data; stationary LoRa gateways that will receive wireless transmissions from mobile containers and a stationary backend network server for monitoring where data is analyzed for fault detection. Sensor nodes collect temperature data and send it to gateways via LoRaWAN communication. Gateways forward the data to the server for processing. As depicted in the technical illustration, the backend segment of the architecture transition from LoRa-based wireless links to fixed TCP/IP connections, ensuring that processed data from the Cloud Server is reliably delivered to the Operator Dashboard for real-time monitoring.

III. Simulation Methodology and Modelling

The system is modelled as mobile sensor nodes, gateways, and a backend server. Sensor nodes generate periodic data using random distributions and backend server evaluates the data to generate event-driven alerts if treshold values are crossed.

The simulation considers communication delays caused by LoRa, packet loss, and queueing effects at the gateway.

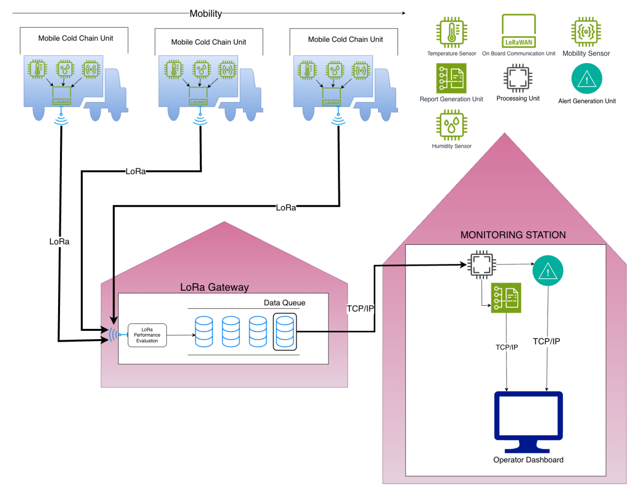

Fig. 1. Cold-chain monitoring system architecture

Objectives:

- Utilize random number generation in data generation
- Evaluate packet delivery ratio
- Measure end-to-end delay
- Analyse queueing behaviour
- Study system performance under mobility

IV. Detailed Plan for Implementing Course Topics

A. Simulation Methodology and Modelling

The system is modelled using discrete-event simulation to represent the logistics chain where events such as packet generation and transmission define system behaviour. Sensors transmit data packets at regular intervals; if these packets exceed defined thresholds, an alert is triggered, transitioning the system to an active state. The system automatically reverts to its baseline state once subsequent readings fall within acceptable limits.

B. Random Number Generation

Random number generation will be utilized for representing temperature, humidity and mobility readings from the sensors, traffic generation, and for the probability models that will be utilized such as Exponential Distribution for Inter-Arrival times (λ), Uniform Distribution for report offsets and Normal

Distribution for temperature variation (μ, σ). By utilizing pseudo-random sequences, researchers can emulate the unpredictable nature of sensor readings and network traffic, ensuring that simulation outcomes are not merely deterministic but represent a statistically valid range of real-world scenarios [6].

C. Monte Carlo Simulation

Monte Carlo Simulation repeatedly runs using different seeds for each parameter configuration, applying random sampling from defined probability distributions to simulate a wide range of potential outcomes [1]. This ensures that analysis of "alarm delay" is statistically significant and accounts for the randomness in wireless contention.

D. Probability Models

Exponential distribution will be employed to model the Inter-Arrival Time(λ) of urgent alarm events [4] and for failure of nodes(λ) meanwhile Normal Distributions represent the modeling of temperature variation(μ, σ) [2].

E. Inverse Transformation or Equivalent Sampling

Non-uniform random inputs for specific sensor failure intervals will be generated via Inverse Transformation [3]. The reporting activity of the sensors and the occurrence of temperature spikes will be modeled as Poisson(λ) processes [7].

F. Stochastic Processes

The dynamics of the cold-chain environment are modeled as stochastic processes to capture real-world uncertainty:

- Arrival Processes(λ): The transmission of routine telemetry data is modeled as a periodic process with a random offset, while the occurrence of critical temperature violations is modeled as a Poisson process.
- Failure Processes(λ): Sensor malfunctions and intermittent wireless link failures are represented as stochastic state transitions.
- Environmental Variation(μ, σ): Fluctuations in container temperature are modeled as a continuous stochastic process to trigger threshold-based alarm events realistically.

G. Queueing Analysis

A queueing point will be modeled at the LoRa Gateway, where incoming packets from multiple containers are buffered before being forwarded to the backend where the service time is determined by the Time-on-Air (ToA) of the LoRa packets. We will use M/M/1-style reasoning to interpret the service delay and buffer overflows, due to the M/M/1 queuing system model giving the most optimistic estimate [8]. This estimate can be used as a lower bound for the message loss probability for a given buffer capacity and a channel load factor [8].

H. Discrete-Event Simulation

The project is implemented as a discrete-event simulation (DES) due to the system state only changing at specific points in time when an event occurs.

- Event-Driven Execution: The simulation progresses from one event (e.g., packet transmission, timeout, or sensor reading) to the next, skipping the continuous time intervals where no changes occur.
- State Changes: The system state (e.g., "Gateway Buffer Occupied," "Container Temperature Critical," or "Channel Busy") remains constant between events, making DES more computationally efficient because it focuses on state changes that occur instantaneously at distinct points in time [5] than time-stepped modeling for large-scale IoT networks.

I. Event Scheduling

Events include periodic updates as packet generation, urgent temperature alerts as alert triggering, handling lost packets in the LoRaWAN layer as retransmission, and processing.

J. Time-Stepped versus Trace-Driven Perspective

The simulation is primarily discrete-event based due to alerts being sent on only urgent situations. In the other hand, using trace-driven inputs for the mobility paths of the containers ensures realistic signal variation.

K. Wireless Network Modelling Fundamentals

The simulation utilizes LoRaWAN communication, incorporating fundamental wireless modeling aspects such as realistic network entities and gateway access considerations. We establish technical assumptions regarding communication range, wireless interference between multiple containers, and the bandwidth constraints inherent to the FLoRa framework. The model accounts for the specific device roles within the LoRa architecture to ensure the simulation accurately reflects the physical limitations of a wide-area IoT deployment.

L. OMNeT++ Implementation

OMNeT++ will be used as the core simulation engine and IDE to build simulation modules for sensors, nodes that contain sensors, gateways, and servers.

M. INET Integration

The INET framework is utilized to model the backend communication between the LoRa gateway and the monitoring station.

Communication between the LoRa gateway and the monitoring station is implemented using TCP/IP protocols provided by INET. The monitoring station is modelled using the *StandardHost* module, incorporating protocol stack components from the *inet.networklayer* and *inet.transportlayer* libraries, including IPv4/IPv6 addressing and TCP communication.

The LoRa gateway is connected to the IP-based network and forwards received sensor data to the monitoring station over a TCP connection. Application-level communication is implemented using modules from the *inet.applications* package, enabling data processing, storage, and alert generation within the monitoring station.

N. FLoRa Integration

FLoRa will be used for LoRaWAN simulation including device behaviour and transmission and will be utilized to implement LoRa physical layer, allowing to model the Spreading Factor (SF) allocation.

O. Input Analysis

Input parameters will be justified using realistic assumptions and statistical fitting against chosen probability distributions and sample data. We will use histograms and QQ-plots to verify that our generated temperature data fits the assumed Normal Distribution (μ, σ).

P. Output Analysis

Results will be analysed using mean, variance, and confidence intervals for the end-to-end delay of emergency messages. Basic visualizations such as histograms and bar charts will also be used to present and compare the results.

V. Proposed Performance Metrics

- Packet Delivery Ratio: Success rate of routine wireless transmissions.
- End-to-End Delay: Delay of wireless communication link between the LoRa Performance Evaluation node and the mobile nodes.
- Alert Response Time: Time elapsed from an alert generation to backend reception.
- Queue Length: The amount of alerts the gateway can hold.
- Throughput: The total amount of successful data (telemetry and alarms) received by the monitoring center per unit of time.

VI. Implementation Plan

The technical implementation involves the development of modular components within OMNeT++ to represent refrigerated container sensor nodes, LoRa gateways, and backend monitoring systems. The team will define the application logic for both periodic telemetry and event-driven alarms, ensuring the message flow correctly reflects a real-world logistics chain. To assess the system's robustness, the simulation will evaluate different comparison scenarios, such as varying node densities and fluctuating traffic rates, to observe how the network performs under increasing load.

VII. Expected Challenges and Initial Solutions

Several technical challenges are anticipated, particularly regarding mobility modeling for containers and network congestion at the gateways. Modeling realistic traffic that captures both routine updates and sudden alarm bursts requires careful parameter selection. These challenges will be addressed through initial solutions such as simplified mobility assumptions, rigorous parameter tuning within the FLoRa framework, and the use of adaptive reporting policies to mitigate wireless contention.

VIII. Preliminary Work Plan

The project tasks are divided among team members to ensure progress across modeling, implementation, and statistical analysis. Initial milestones focus on the design of the network topology and the integration of INET and FLoRa modules. Subsequent phases include the implementation of stochastic arrival processes and the execution of Monte Carlo runs, followed by a final period dedicated to the interpretation of output metrics and confidence intervals.

IX. Conclusion

This project provides a technically rigorous simulation-based approach to analyzing cold-chain logistics monitoring systems using OMNeT++. By modeling the interaction between mobile sensors and LoRaWAN infrastructure, the study aims to evaluate system reliability, packet delivery performance, and response times under realistic conditions. The final results will demonstrate how discrete-event simulation can be used to optimize IoT-oriented logistics and ensure the safety of sensitive cargo.

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